

STT 861: Theory of Probability & Statistics I

Lecture 7: Random Variables & Distribution Functions

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Reference: Casella & Berger Ch. 1.4–1.5

Lecture 7 Outline

Today

- Random variables: formal definition
- Cumulative distribution function (CDF)
- Properties of CDF; discrete vs. continuous RVs
- Identically distributed

Why Random Variables?

Motivation

In many experiments it is easier to work with a **summary variable** than with the original probability structure.

Example: Poll 50 people (agree = “1”, disagree = “0”).

- Original sample space: all binary strings of length 50 $\Rightarrow 2^{50}$ elements.
- We only care about X = number of 1s recorded.
- New sample space for X : $\mathcal{X} = \{0, 1, 2, \dots, 50\}$ — much smaller!

What we are really doing

By defining X , we create a **mapping** (a function) from the original sample space to a new, simpler one — usually a set of real numbers.

$$S \xrightarrow{X} \mathcal{X} \subseteq \mathbb{R}$$

This is the idea behind a **random variable**.

Definition 1.4.1 — Random Variable

Definition 1.4.1 (CB)

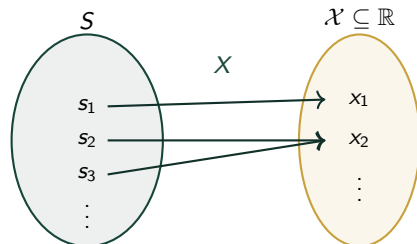
A **random variable** is a function from a sample space S into the real numbers $\mathbb{R}$.

$$X : S \longrightarrow \mathbb{R}$$

The set of values X can take is called the **range** of X , denoted $\mathcal{X}$.

Notation convention

- **Upper case** X : the random variable (the function).
- **Lower case** x : a realized value (a number).
- We write $P(X = x)$ as shorthand for $P(\{s \in S : X(s) = x\})$.



Example 1.4.2 — Random Variables in Practice

Example 1.4.2 (CB): Examples of random variables

Experiment	Random variable X
Toss two dice	$X = \text{sum of the two numbers}$
Toss a coin 25 times	$X = \text{number of heads in 25 tosses}$
Apply different amounts of fertilizer to corn plants	$X = \text{yield per acre}$

Key point

In each case, X collapses a complicated sample space into a single number that captures the quantity of scientific interest. We can then work with X directly, without reference to the original S .

Induced Probability Function

Induced Probability (CB Eq. 1.4.1)

Let $S = \{s_1, \dots, s_n\}$ with probability function P , and let X be a random variable with range $\mathcal{X} = \{x_1, \dots, x_m\}$.

The **induced probability function** P_X on $\mathcal{X}$ is:

$$P_X(X = x_i) = P(\{s_j \in S : X(s_j) = x_i\}) \quad (1.4.1)$$

For any set $A \subseteq \mathcal{X}$:

$$P_X(X \in A) = P(\{s \in S : X(s) \in A\}) \quad (1.4.2)$$

Notation simplification

Because of the equivalence in (1.4.1), we simply write $P(X = x_i)$ rather than $P_X(X = x_i)$.

P_X satisfies the Kolmogorov Axioms (see Exercise 1.45), so it is a valid probability function on $\mathcal{X}$.

Example 1.4.3 — Three Coin Tosses II

Example 1.4.3 (CB): Three coin tosses

From Example 1.3.13: $S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$, each with probability $\frac{1}{8}$.

Define $X =$ number of heads.

Value of X for each sample point:

s	HHH	HHT	HTH	THH	TTH	THT	HTT	TTT
$X(s)$	3	2	2	2	1	1	1	0

Induced probability function:

x	$P_X(X = x)$
0	$\frac{1}{8}$
1	$\frac{3}{8}$
2	$\frac{3}{8}$
3	$\frac{1}{8}$

Example 1.4.4 — Large Sample Space

Example 1.4.4 (CB): Distribution of a random variable

Poll 50 people. S = all binary strings of length 50 (2^{50} elements).
 X = number of 1s (agrees), $\mathcal{X} = \{0, 1, 2, \dots, 50\}$.

Assume each string equally likely $\Rightarrow P(\text{each string}) = \frac{1}{2^{50}}$.

How many strings have exactly 27 ones? $\binom{50}{27}$.

Therefore:

$$P_X(X = 27) = \frac{\binom{50}{27}}{2^{50}}$$

In general, for any $i \in \mathcal{X}$:

$$P_X(X = i) = \frac{\binom{50}{i}}{2^{50}}$$

Remark

Even though $|S| = 2^{50}$ is enormous, the induced probability function P_X is determined by a simple formula. This is the power of random variables.

Extension to Uncountable Range

General Induced Probability (Eq. 1.4.2)

If $\mathcal{X}$ is uncountable (e.g. $\mathcal{X} = \mathbb{R}$ or $\mathcal{X} = [0, \infty)$), we cannot list individual values. Instead, for any set $A \subset \mathcal{X}$:

$$P_X(X \in A) = P(\{s \in S : X(s) \in A\})$$

This defines a valid probability function on $\mathcal{X}$, satisfying the Kolmogorov Axioms.

Examples with uncountable $\mathcal{X}$

- Survival time after surgery: $X \geq 0$, $\mathcal{X} = [0, \infty)$
- Blood pressure measurement: $\mathcal{X} = \mathbb{R}$
- Yield per acre (fertilizer example): $\mathcal{X} = (0, \infty)$

In each case we assign probabilities to *intervals* such as $P(a < X \leq b)$, not to individual points.

Definition 1.5.1 — Cumulative Distribution Function

Definition 1.5.1 (CB)

With every random variable X we associate a function called the **cumulative distribution function** (cdf) of X , denoted $F_X(x)$:

$$F_X(x) = P_X(X \leq x), \quad \text{for all } x \in \mathbb{R}.$$

Key features:

- F_X is defined for *all* $x \in \mathbb{R}$, not just the values in $\mathcal{X}$.
- $F_X(x)$ gives the probability that X takes a value *at or below* x .
- F_X completely determines the probability distribution of X .

Useful formulas

For any $a < b$:

$$P(a < X \leq b) = F_X(b) - F_X(a)$$

$$P(X > a) = 1 - F_X(a)$$

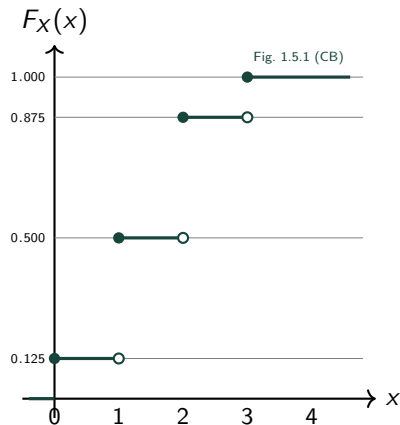
Example 1.5.2 — CDF of Three Coin Tosses

X = number of heads in 3 fair coin tosses.

The cdf is:

$$F_X(x) = \begin{cases} 0 & -\infty < x < 0 \\ \frac{1}{8} & 0 \leq x < 1 \\ \frac{1}{2} & 1 \leq x < 2 \\ \frac{7}{8} & 2 \leq x < 3 \\ 1 & 3 \leq x < \infty \end{cases}$$

E.g. $F_X(2.5) = P(X \leq 2.5) = \frac{7}{8}$.



Theorem 1.5.3 — Characterisation of CDFs

Theorem 1.5.3 (CB)

A function $F(x)$ is a cdf if and only if:

- (a) $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow +\infty} F(x) = 1$.
- (b) $F(x)$ is **nondecreasing** in x .
- (c) $F(x)$ is **right-continuous**: for every x_0 , $\lim_{x \downarrow x_0} F(x) = F(x_0)$.

Remarks on right-continuity

- At a jump point x_i , F_X takes the value at the *top* of the jump (hence the $\leq$ in $P(X \leq x)$, not $<$).
- The **size of the jump** at x equals $P(X = x)$.
- Defining $F_X(x) = P(X < x)$ (strict inequality) would give a *left-continuous* function instead.

Example 1.5.4 — Geometric CDF (Discrete)

Example 1.5.4 (CB): Tossing for a head

Toss a coin ($P(\text{head}) = p$) until a head appears. X = number of tosses. For $x = 1, 2, \dots$:

$$P(X = x) = (1 - p)^{x-1}p$$

Using the geometric series $\sum_{k=1}^n t^{k-1} = \frac{1 - t^n}{1 - t}$:

$$F_X(x) = \sum_{i=1}^x (1 - p)^{i-1}p = p \cdot \frac{1 - (1 - p)^x}{p} = 1 - (1 - p)^x, \quad x = 1, 2, \dots$$

Verification of Theorem 1.5.3

(a) $F_X(x) = 0$ for $x < 0$; $\lim_{x \rightarrow \infty} [1 - (1 - p)^x] = 1$.✓ **(b)** $(1 - p)^x$ decreases in x , so F_X increases.✓ **(c)** $F_X(x + \varepsilon) = F_X(x)$ for small $\varepsilon > 0$.✓

F_X is the cdf of the **geometric distribution** (Fig. 1.5.2, CB).

Example 1.5.5 — Logistic CDF (Continuous)

Example 1.5.5 (CB): Logistic cdf

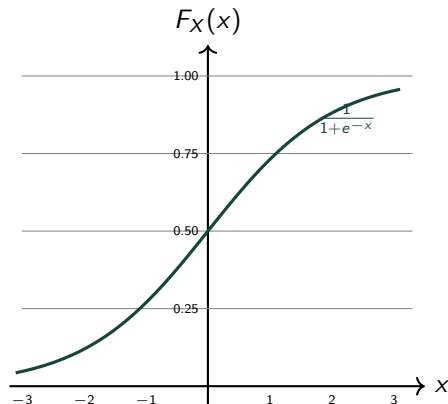
$$F_X(x) = \frac{1}{1 + e^{-x}}, \quad x \in \mathbb{R}$$

(a) $\lim_{x \rightarrow -\infty} F_X(x) = 0$, $\lim_{x \rightarrow \infty} F_X(x) = 1$.✓

(b) $\frac{d}{dx} F_X(x) = \frac{e^{-x}}{(1 + e^{-x})^2} > 0$.✓

(c) F_X is continuous everywhere.✓

The derivative $f_X(x) = \frac{e^{-x}}{(1 + e^{-x})^2}$ is the **probability density function** (pdf) of the logistic distribution.



Example 1.5.6 — CDF With a Jump (Mixed Distribution)

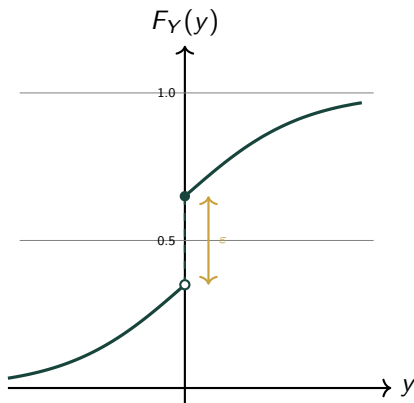
Example 1.5.6 (CB): Gauge that sticks at 0

Modify the logistic cdf: a gauge gives a continuous reading, but sticks at $y = 0$ with probability ε ($0 < \varepsilon < 1$):

$$F_Y(y) = \begin{cases} \frac{1 - \varepsilon}{1 + e^{-y}} & y < 0 \\ \varepsilon + \frac{1 - \varepsilon}{1 + e^{-y}} & y \geq 0 \end{cases}$$

Structure:

- Continuous for $y < 0$ and $y > 0$.
- **Jump of size ε** at $y = 0$, i.e. $P(Y = 0) = \varepsilon$.
- Mixed distribution: point mass at 0, continuous elsewhere.



Definition 1.5.7 & 1.5.8 — Types and Identical Distribution

Definition 1.5.7 (CB): Continuous and Discrete

- X is **continuous** if $F_X(x)$ is a continuous function of x .
- X is **discrete** if $F_X(x)$ is a step function of x .

Definition 1.5.8 (CB): Identically Distributed

X and Y are **identically distributed** if for every $A \in \mathcal{B}^1$:

$$P(X \in A) = P(Y \in A)$$

Theorem 1.5.10 (CB)

The following are equivalent: **(a)** X and Y are identically distributed. **(b)** $F_X(x) = F_Y(x)$ for every $x \in \mathbb{R}$.

Section 1.5 — Summary

	Discrete	Continuous
CDF shape	Step function	Smooth, differentiable
Jump at x	$P(X = x)$	0 (no point masses)
CB examples	Ex 1.5.2 (coins), 1.5.4 (geometric)	Ex 1.5.5 (logistic)
Right-continuous	Yes (Thm 1.5.3c)	Yes (trivially)

Key Results of §1.5

- ① **Def. 1.5.1:** $F_X(x) = P(X \leq x)$ for all $x \in \mathbb{R}$.
- ② **Thm 1.5.3:** F is a cdf $\iff$ limits $(0, 1)$, nondecreasing, right-continuous.
- ③ **Def. 1.5.7:** Continuous vs discrete RV via shape of F_X .
- ④ **Thm 1.5.10:** $X \stackrel{d}{=} Y \iff F_X = F_Y$.

Activity 7 – CDF

Activity 7

- Ⓐ Roll a fair die. Write down $F_X(x)$ for all $x \in \mathbb{R}$ and sketch the graph.
- Ⓑ A RV X has CDF $F(x) = 0$ for $x < 0$, $F(x) = x^2$ for $0 \leq x \leq 1$, $F(x) = 1$ for $x > 1$.
 - Ⓐ Verify the three CDF properties.
 - Ⓑ Find $P(0.25 < X \leq 0.5)$.
 - Ⓒ Find $P(X = 0.5)$.
- Ⓒ **Bonus:** What type of RV is (a)? What type is (b)?

Activity 7 – Solutions

- Ⓐ $F(x) = 0$ for $x < 1$; $F(x) = k/6$ for $k \leq x < k + 1$, $k = 1, \dots, 5$; $F(x) = 1$ for $x \geq 6$. Staircase function.
- Ⓑ (i) $F(0^-) = 0$ ✓; $F(\infty) = 1$ ✓; F is non-decreasing✓; right-continuous✓.
(ii) $F(0.5) - F(0.25) = 0.25 - 0.0625 = \mathbf{0.1875}$
(iii) $P(X = 0.5) = F(0.5) - F(0.5^-) = 0.25 - 0.25 = \mathbf{0}$ (continuous RV)
- Ⓒ (a) Discrete; (b) Continuous.